

29 October.

CNN (Convolution Neural Networks)

It is type of DNN which process data in grid like ~~technology~~ ^{topology}. It is good for pattern recognition.

- 1) Image classification. (eg cars, dogs)
 - 2) Object Detection. (Number plates, humans in crowd)
 - 3) Face ^{Recognition} ~~Recognition~~
 - 4) Medical imaging (x-rays) during covid times they did xray and used CNN to identify
 - 5) Self driving cars
 - 6) OCR - written text.
- ### Architecture.

- 1) INPUT LAYER: Take image input. (28x28) ^{pixels}
(764) inputs _{mnist Data samples.} ^{766 pixels.}

Mnist is for digits
fashion mnist for clothes.
in IRIS there were 4 inputs

- 2) CONVOLUTION LAYER: Applies kernel to extract the features like edges, colors & textures etc.
- 3) Activation function: Relu, $\max(0, x)$
- 4) Pooling layer: Reduces spatial size.
When we apply kernels and combine the numeric array then we can have a better understanding of the pattern. Pooling.